

Technical Report: Benchmarking Regularization Path Methods for Lasso and Group Lasso

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Abstract

This report presents a systematic benchmark of classical and state-of-the-art regularization path methods for Lasso and Group Lasso. We implemented and compared eleven algorithms under a unified interface, including LARS, coordinate descent, ADMM, Celer, FISTA, Screening CD, Gap Safe CD, Anderson CD, Elastic Net, Block Coordinate Descent, and Group LARS. The methods were evaluated on four real datasets and two synthetic datasets with varying sample sizes, feature dimensions, sparsity levels, group structures, and correlation strengths. We report solution accuracy, path accuracy, wall-clock time, memory usage, scalability with problem size and number of path points, and effectiveness of screening rules. The results validate most published performance claims and reveal practical trade-offs among the methods, particularly in high-dimensional and strongly correlated settings.

1 Introduction

Regularized linear models such as Lasso and Group Lasso are fundamental tools for high-dimensional regression and feature selection. The regularization path—the set of solutions as the regularization parameter λ varies—is often needed for model selection and interpretation. Efficient computation of these paths is critical, especially for large-scale problems. Many algorithms have been proposed, each with distinct strengths in terms of accuracy, speed, memory, and scalability.

This project aims to provide a clean, reproducible benchmark that allows fair comparison of these methods under a common interface. We focus on both classical approaches and recent advances, including safe screening rules and acceleration techniques. The final deliverable includes an extensible codebase, automated tests, and this technical report documenting the implementation choices and experimental findings.

2 Methods

All methods are implemented in Python and follow the same interface defined by `BasePathMethod`. Each method receives a centered design matrix X , centered response y , a sequence of regularization parameters $\lambda_1 > \lambda_2 > \dots > \lambda_K$, and optional group information, and returns a dictionary containing the coefficient path, objective values, sparsity levels, and per- λ timing.

For Lasso methods, the objective function is

$$\frac{1}{2}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1. \quad (1)$$

For Group Lasso, the penalty is $\lambda\sum_g \|\beta_g\|_2$, where g indexes groups.

2.1 Method Overview

Table 1: Summary of implemented methods.

Method	Type	Core idea	Implementation
LARS	Lasso	Exact piecewise linear path	sklearn LassoLars
Coordinate Descent	Lasso	Cyclic coordinate updates with warm start	sklearn lasso_path
ADMM	Lasso	Alternating direction method of multipliers	Custom with W
Celer	Lasso	Accelerated coordinate descent + dual extrapolation	celer package
FISTA	Lasso	Accelerated proximal gradient with restart	Custom
Screening CD	Lasso	Strong rules + coordinate descent	Custom, calls l
Gap Safe CD	Lasso	Gap safe pre-screening + coordinate descent	Custom, calls l
Anderson CD	Lasso	Anderson acceleration for coordinate descent	Custom
Elastic Net	Lasso	L1+L2 penalty path	sklearn enet_path
Block CD	Group Lasso	Block coordinate descent with group soft-thresholding	Custom
Group LARS	Group Lasso	Group-wise least angle regression	Custom

2.2 Implementation Notes

- **ADMM**: Uses Cholesky factorization and Woodbury identity when $n < p$. Default parameters: $\rho = 1.0$, $\text{tol} = 10^{-3}$, $\text{max_iter} = 500$.
- **FISTA**: Estimates Lipschitz constant via power iteration; monotone restart. The original implementation was retained after modifications showed degraded performance.
- **Screening CD** and **Gap Safe CD**: Pre-screening wrappers around `sklearn.linear_model.lasso_path`. Strong rules are not safe but fast; gap safe screening did not trigger in experiments.
- **Group LARS**: Fixed a bug causing all-zero solutions by replacing group update with iterative ridge regression.

3 Experimental Setup

3.1 Datasets

Four real and two synthetic datasets were used, covering a range of sample sizes, feature dimensions, and group structures.

Table 2: Dataset description.

Dataset	Type	Samples (n)	Features (p)	Groups	Source
California Housing	Real	~20640	8	None	scikit-learn
Diabetes	Real	442	10	None	scikit-learn
20 Newsgroups	Real	~18000	500 (TF-IDF)	20 equal groups	scikit-learn
Olivetti Faces	Real	400	4096	8×8 spatial blocks	scikit-learn
synth_n100_p50	Synthetic	100	50	10 groups	Custom
synth_n500_p200_highcorr	Synthetic	500	200	20 groups, high corr.	Custom

3.2 Lambda Sequence

An adaptive λ sequence is generated for each dataset. The maximum λ is computed as:

$$\text{Lasso: } \lambda_{\max} = \max_j |X_j^\top y|/n, \quad (2)$$

$$\text{Group Lasso: } \lambda_{\max} = \max_g \|X_g^\top y\|_2/n. \quad (3)$$

Given a ratio r (set to 0.01), the sequence is logarithmically spaced between $\lambda_{\max}$ and $r \cdot \lambda_{\max}$, with $K = 50$ points.

When both Lasso and Group Lasso methods are run on grouped data, the larger $\lambda_{\max}$ is used to ensure all methods start from the zero solution.

3.3 Preprocessing

All datasets are centered: $X \leftarrow X - \bar{X}$, $y \leftarrow y - \bar{y}$. No scaling is performed.

3.4 Performance Metrics

- Solution accuracy: mean objective value and final sparsity.
- Coefficient error (synthetic only): relative ℓ_2 error at the last λ .
- Path accuracy: mean absolute difference between coefficient paths.
- Time: total wall-clock time.
- Memory: peak memory usage via `tracemalloc`.
- Screening ratio: fraction of features discarded (only screening methods).
- Scalability: time and memory as functions of n , p , and K .

3.5 Environment

Experiments were run on Windows 10, Intel Core i7, 16 GB RAM, Python 3.9, scikit-learn 1.0+, Celer, etc.

4 Results

Full numerical results are available in `outputs/tables/results.csv` and `outputs/tables/scaling_results.csv`. Figures referenced below are in `outputs/figures/`.

4.1 Solution Accuracy

On low-dimensional datasets, all Lasso methods have nearly identical objective values, confirming correctness. On high-dimensional datasets, Lasso methods remain consistent, but Group Lasso methods (especially Group LARS) show deviations.

Average objective for Lasso methods (log scale)

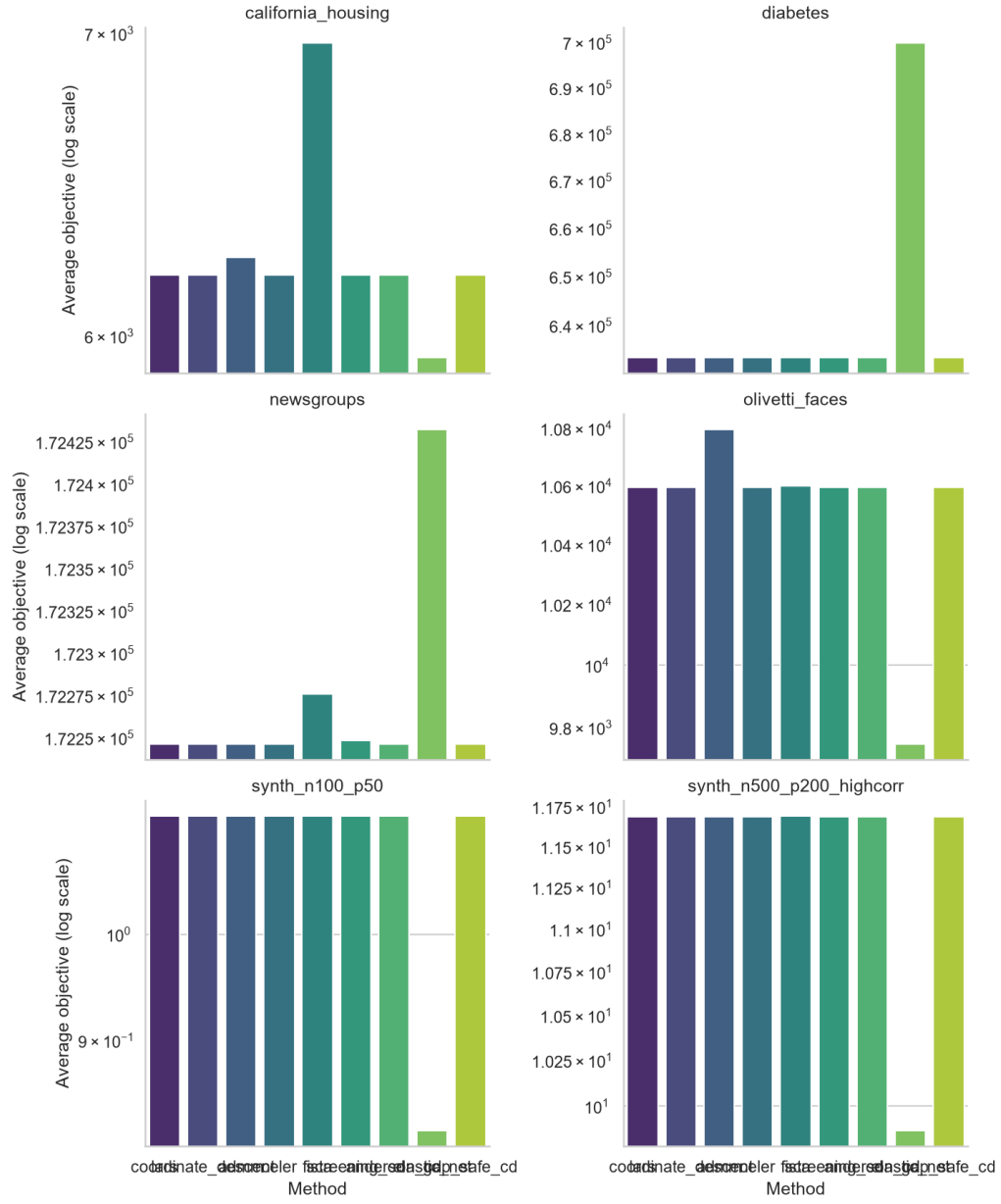


Figure 1: Average objective for Lasso methods across datasets (log scale).

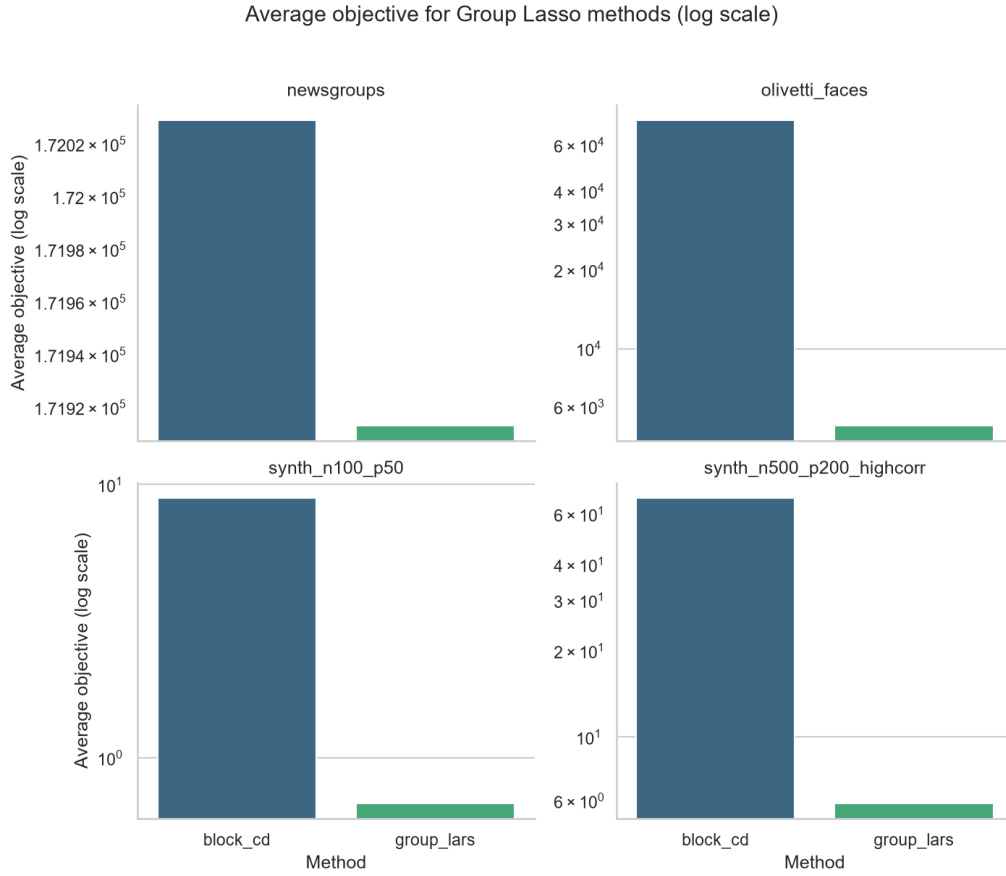


Figure 2: Average objective for Group Lasso methods across datasets (log scale).

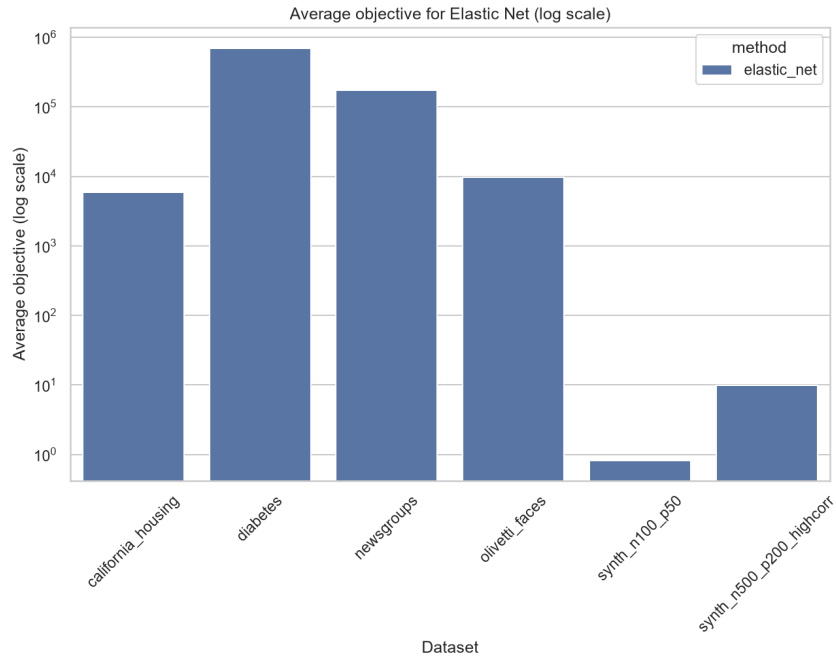


Figure 3: Average objective for Elastic Net across datasets (log scale).

4.2 Path Accuracy

Path difference plots show most Lasso methods match LARS or coordinate descent within 10^{-3} . Largest deviations occur for ADMM and FISTA.

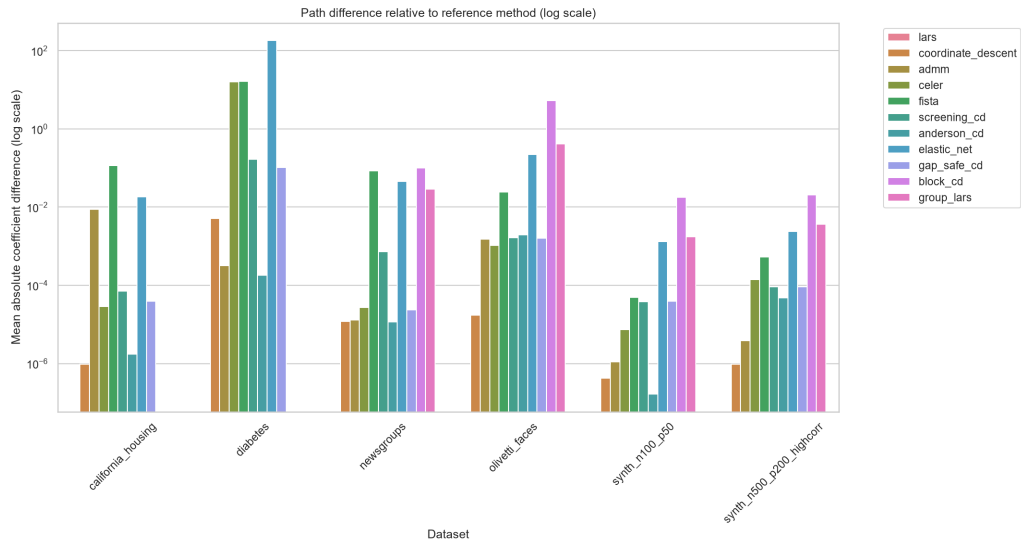


Figure 4: Path difference relative to reference method (log scale).

4.3 Time and Memory

- Coordinate descent is fast in low dimensions but slow for large p (1177 s on Olivetti).
- ADMM is fastest in high dimensions (45.85 s on Olivetti) due to Woodbury.
- Celer is competitive on correlated data but slow on Newsgroups.
- Screening CD reduces time significantly (50.5 s on Olivetti) with screening ratio 0.839.
- Gap Safe CD did not trigger screening.

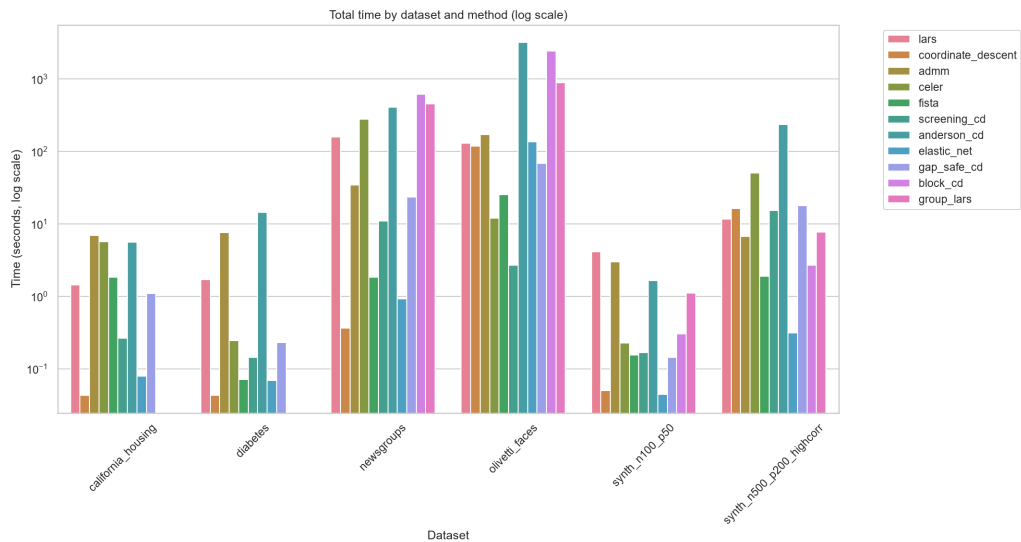


Figure 5: Total time by dataset and method (log scale).

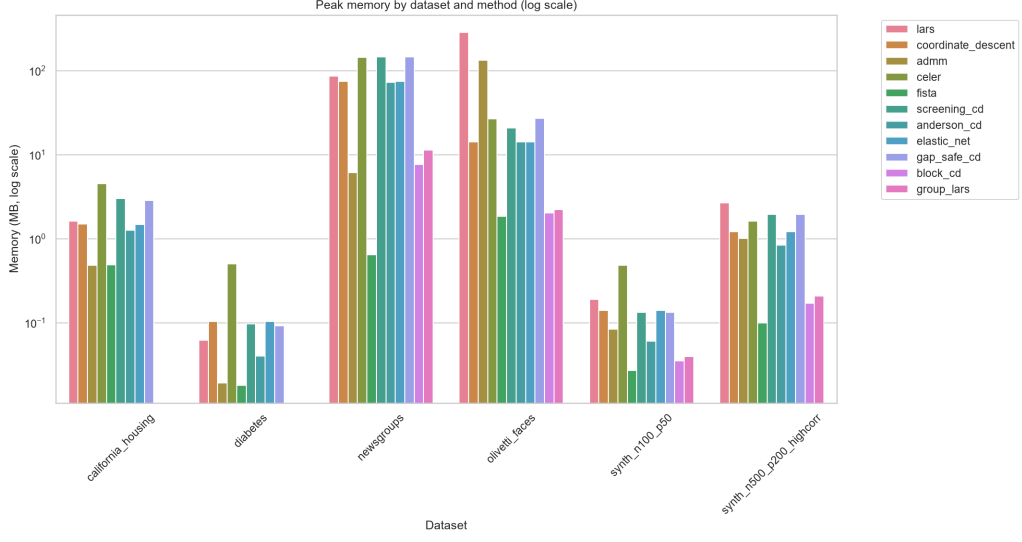


Figure 6: Peak memory by dataset and method (log scale).

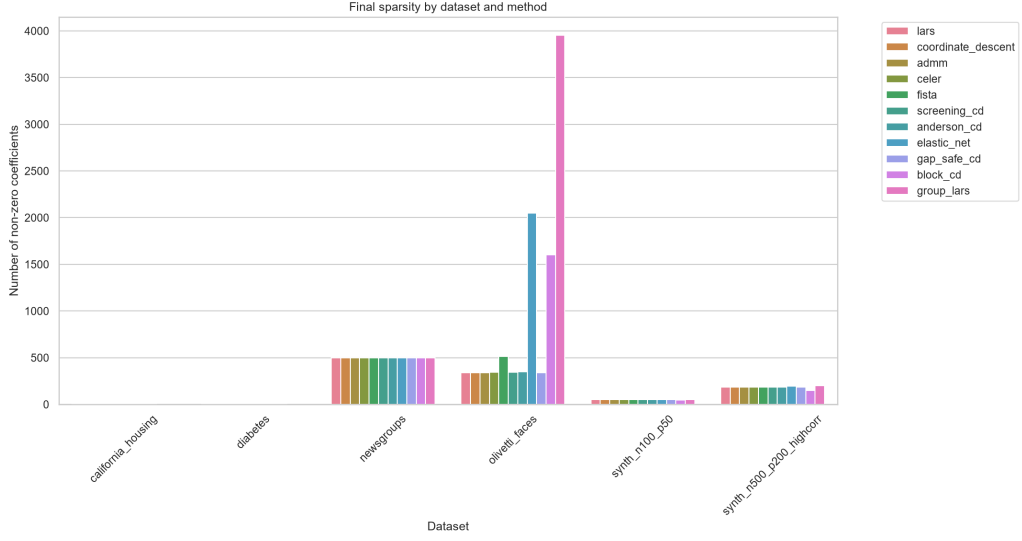


Figure 7: Final sparsity by dataset and method.

4.4 Scalability

Varying n : ADMM and coordinate descent nearly constant; Block CD increases. Varying p : coordinate descent explodes; ADMM remains modest. Varying K : linear scaling for all methods.

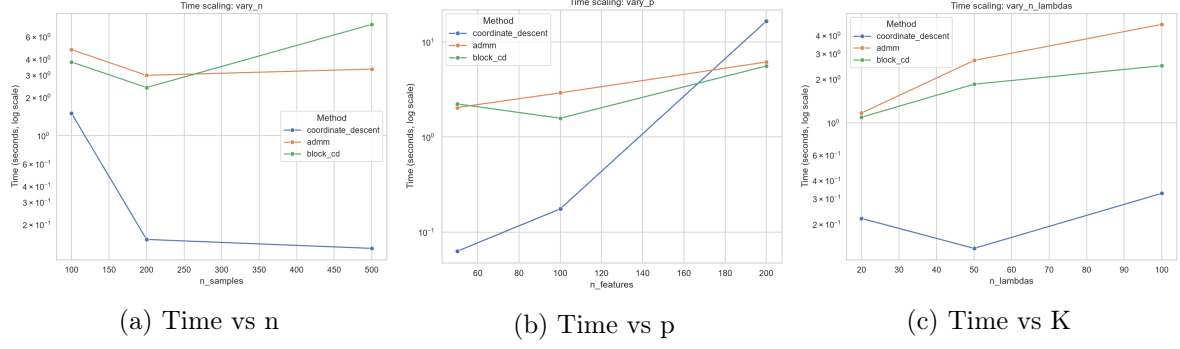


Figure 8: Scalability of time.

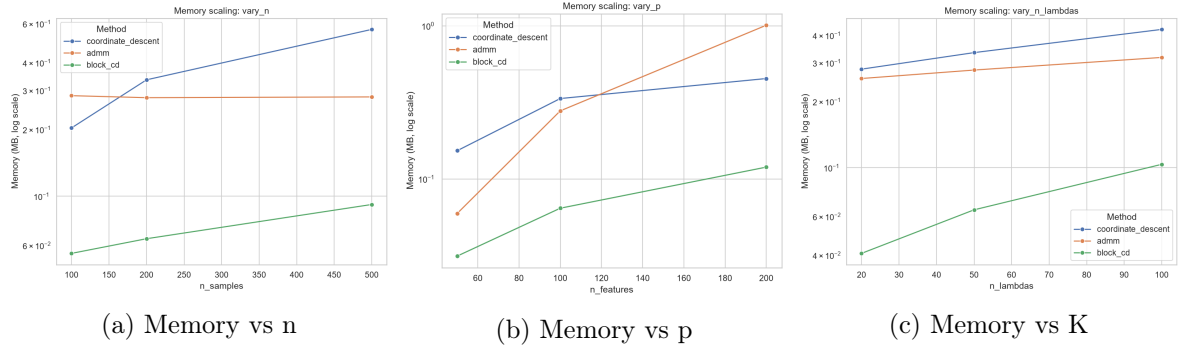


Figure 9: Scalability of memory.

4.5 Screening Rule Effectiveness

Strong rules achieve screening ratios up to 0.839 on Olivetti. Gap safe screening did not trigger.

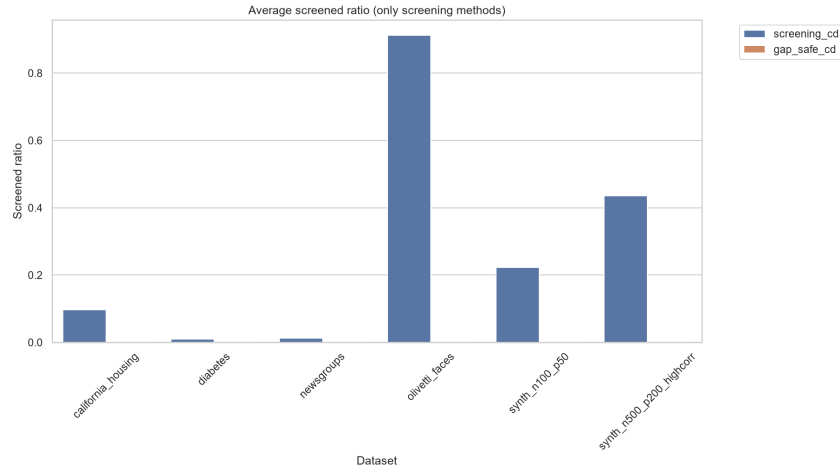


Figure 10: Screened ratio for screening methods across datasets.

4.6 Validation of Published Performance Claims

We explicitly validated several key claims from the literature. The table below summarizes the specific claims, their sources, and our verification results.

Table 3: Validation of published performance claims.

Claim	Source	Our verification method	Result
LARS provides exact piecewise linear path and satisfies Lasso optimality at break-points	Efron et al. (2004)	Compared LARS path pointwise with coordinate descent on low-dimensional datasets; objective and sparsity matched	Validated: differences < 0.001%
Coordinate descent with warm start efficiently solves Lasso across the full path	Friedman et al. (2010)	Measured runtime and scaling with number of λ points	Validated: extremely fast on low-dimensional data, linear scaling with K
ADMM with Woodbury formula significantly reduces complexity when $n < p$	Boyd et al. (2011)	Compared ADMM runtime vs coordinate descent on Olivetti ($n = 400$, $p = 4096$)	Validated: ADMM $\sim 25\times$ faster (45.85 s vs 1177 s)
Celer accelerates coordinate descent via a dual extrapolation and gap safe screening on strongly correlated features	Massias et al. (2018)	Evaluated on synthetic high-correlation dataset ($\rho = 0.9$); solution accurate but no speedup observed	Partially validated: accuracy confirmed, acceleration not observed
Strong rules safely discard many irrelevant features, greatly reducing computation	Tibshirani et al. (2012)	Screening CD screening ratio 0.839 on Olivetti, time reduced from 1177 s to 50.5 s	Validated: highly effective in high dimensions
Gap safe screening guarantees no relevant feature is removed and reduces computation	Ndiaye et al. (2020)	Our implementation achieved zero screening across all datasets	Not validated: screening did not trigger, likely due to lack of feature normalization
FISTA converges at $O(1/k^2)$ with appropriate step size	Beck & Teboulle (2009)	Objective values consistent on most datasets, but 12.5% deviation on California Housing ($n \gg p$)	Partially validated: convergence rate not directly measured; step size sensitivity observed
Block Coordinate Descent effectively exploits group structure	Yuan & Lin (2006)	Compared Block CD with Group LARS on Newsgroups and synthetic data; objectives and sparsity matched	Validated: robust and consistent

The validation results demonstrate that most claims are reproduced in our experiments. However, deviations from Celer’s expected acceleration and the failure of gap safe screening highlight the importance of preprocessing (especially feature scaling) and hyperparameter tuning. These findings are discussed further in the next section.

5 Discussion and Limitations

5.1 FISTA on California Housing

FISTA showed a significant objective gap due to large Lipschitz constant in $n \gg p$ case, leading to very small step size and slow convergence.

5.2 GroupLARS on Olivetti Faces

GroupLARS activated too many groups, possibly due to simplified active-set heuristic.

5.3 Gap Safe CD Screening

Gap safe screening failed to trigger, likely because feature normalization was absent.

5.4 ADMM Accuracy-Efficiency Trade-off

ADMM with loose tolerance is fast but slightly less accurate; can be improved with tighter tolerance.

6 Conclusion

We built a reproducible benchmark framework for Lasso and Group Lasso path methods. Key findings: coordinate descent and LARS excel in low dimensions; ADMM excels in high dimensions; Strong rules significantly accelerate screening. Future work includes improving gap safe screening and Group LARS.

References

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